

Lecture 12

The $\bar{\partial}$ -Poincaré lemma in one variable

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Lecture 11: Dolbeault's theorem

Let M be a complex manifold, Ω_M^p the sheaf of holomorphic p -forms, and $\mathcal{A}_M^{p,q}$ the sheaf of smooth (p, q) -forms.

Theorem.

$$H^q(M, \Omega_M^p) \cong H_{\bar{\partial}}^{p,q}(M) = \frac{\ker(\bar{\partial} : \mathcal{A}_M^{p,q}(M) \rightarrow \mathcal{A}_M^{p,q+1}(M))}{\operatorname{im}(\bar{\partial} : \mathcal{A}_M^{p,q-1}(M) \rightarrow \mathcal{A}_M^{p,q}(M))}.$$

Method. Use the acyclic resolution

$$0 \longrightarrow \Omega_M^p \longrightarrow \mathcal{A}_M^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}_M^{p,1} \xrightarrow{\bar{\partial}} \dots$$

Exactness comes from the $\bar{\partial}$ -Poincaré lemma. Acyclicity comes from partitions of unity.

Verbitsky, Lecture 11, pp. 20, 28; `de-rham-dolbeault` notes, Dolbeault's theorem.

Why an acyclic resolution computes cohomology

Recall. A sheaf $\mathcal{B}$ is acyclic if $H^r(M, \mathcal{B}) = 0$ for $r > 0$.

Theorem. If $0 \rightarrow \mathcal{F} \rightarrow \mathcal{B}^0 \rightarrow \mathcal{B}^1 \rightarrow \dots$ is exact and each $\mathcal{B}^j$ is acyclic, then

$$H^q(M, \mathcal{F}) \cong H^q(\Gamma(M, \mathcal{B}^\bullet)).$$

Proof. Put $Z^0 = \mathcal{F}$ and $Z^j = \ker(\mathcal{B}^j \rightarrow \mathcal{B}^{j+1})$. The sequences $0 \rightarrow Z^j \rightarrow \mathcal{B}^j \rightarrow Z^{j+1} \rightarrow 0$ give

$$\frac{\Gamma(Z^q)}{\operatorname{im} \Gamma(\mathcal{B}^{q-1})} \cong H^1(M, Z^{q-1}) \cong \dots \cong H^q(M, \mathcal{F}) \quad (q \geq 1).$$

These are the long exact cohomology sequences, using acyclicity. For $q = 0$, the kernel is $\Gamma(M, \mathcal{F})$. □

Verbitsky, Lecture 11, pp. 18, 20, 27; `de-rham-dolbeault`, Proposition 4.32.

The Dolbeault resolution: exactness

At degree zero. In holomorphic coordinates,

$$\alpha = \sum_{|J|=p} f_J dz_J, \quad \bar{\partial}\alpha = 0 \quad \Longleftrightarrow \quad \frac{\partial f_J}{\partial \bar{z}_k} = 0 \quad \text{for all } J, k.$$

Thus $\ker(\bar{\partial} : \mathcal{A}_M^{p,0} \rightarrow \mathcal{A}_M^{p,1}) = \Omega_M^p$.

At positive degree. The $\bar{\partial}$ -Poincaré lemma says that if $q > 0$ and $\bar{\partial}\alpha = 0$, then locally $\alpha = \bar{\partial}\beta$.

Exactness of sheaves is exactness on stalks, so local primitives suffice.

This is the input we begin proving today.

Verbitsky, Lecture 11, pp. 8, 28; `de-rham-dolbeault`, Dolbeault proof, steps 1–2.

The Dolbeault resolution: acyclicity

Each $\mathcal{A}_M^{p,q}$ is a sheaf of C_M^∞ -modules, hence fine. Here is why partitions of unity kill its higher cohomology.

Let (ρ_j) be a subordinate partition of unity, and let $c = (c_{i_0 \dots i_r})$ be an alternating Čech cocycle, $r > 0$. Define

$$b_{i_0 \dots i_{r-1}} = \sum_j \rho_j c_{j i_0 \dots i_{r-1}}.$$

Products extend by zero, and the sum is locally finite. Since $\delta c = 0$,

$$(\delta b)_{i_0 \dots i_r} = \sum_j \rho_j \sum_{k=0}^r (-1)^k c_{j i_0 \dots \widehat{i_k} \dots i_r} = \sum_j \rho_j c_{i_0 \dots i_r} = c_{i_0 \dots i_r}.$$

Thus $H^r(M, \mathcal{A}_M^{p,q}) = 0$ for $r > 0$. The acyclic resolution theorem now gives Dolbeault's theorem. □

Verbitsky, Lecture 11, pp. 11–18; `de-rham-dolbeault`, Propositions 4.36 and 4.32.

Today's theorem: one complex variable

The $\bar{\partial}$ -Poincaré lemma (local form). If η is a smooth $(0,1)$ -form near a point of $\mathbb{C}$, then on a smaller disk there is a smooth function u such that

$$\bar{\partial}u = \eta.$$

More precisely, if η is smooth on a neighbourhood of a closed disk $\bar{D}$, we will construct u on D .

For $z = x + iy$,

$$\eta = g d\bar{z}, \quad \bar{\partial}u = \frac{1}{2}(u_x + iu_y) d\bar{z}.$$

Every $(0,1)$ -form is $\bar{\partial}$ -closed: $d\bar{z} \wedge d\bar{z} = 0$. The $(1,1)$ case follows from the same scalar equation.

Next lecture

Several-variable $\bar{\partial}$ -Poincaré lemma. For a smooth (p, q) -form η on a complex manifold,

$$q > 0, \quad \bar{\partial}\eta = 0 \quad \implies \quad \eta = \bar{\partial}\beta \text{ locally,} \quad \beta \text{ of type } (p, q-1).$$

This completes the local exactness needed for Dolbeault's theorem.

Hartogs extension. Let $n \geq 2$ and $K \subset \mathbb{C}^n$ be compact with $\mathbb{C}^n \setminus K$ connected. Every holomorphic function on $\mathbb{C}^n \setminus K$ extends uniquely to a holomorphic function on $\mathbb{C}^n$.

Verbitsky, Lecture 13: Poincaré–Dolbeault–Grothendieck lemma and Hartogs' theorem.

Hilbert spaces: what we need

A complex **Hilbert space** is a vector space with an inner product, complete for the norm $\|v\| = \sqrt{\langle v, v \rangle}$. We take the inner product linear in its first argument.

Our example is $L^2(0, 2\pi]$: square-integrable complex functions, identified when equal almost everywhere, with

$$\langle f, g \rangle = \int_0^{2\pi} f(t) \overline{g(t)} dt, \quad \|f\|_2^2 = \int_0^{2\pi} |f(t)|^2 dt.$$

An **orthonormal system** (e_j) satisfies $\langle e_j, e_k \rangle = \delta_{jk}$. It is an **orthonormal basis** if its finite linear combinations are dense in the Hilbert space.

Here “basis” allows convergent infinite sums.

Why an orthonormal basis gives a series

Theorem. For a countable orthonormal basis (e_j) ,

$$v = \sum_j \langle v, e_j \rangle e_j, \quad \|v\|^2 = \sum_j |\langle v, e_j \rangle|^2.$$

Proof. Write $S_N = \sum_{j=1}^N \langle v, e_j \rangle e_j$. Orthogonality gives

$$\|v - S_N\|^2 = \|v\|^2 - \sum_{j=1}^N |\langle v, e_j \rangle|^2 \geq 0.$$

The numerical sum converges, so (S_N) is Cauchy and has a limit s . For every j , $\langle v - s, e_j \rangle = 0$. Density then gives $v = s$. The norm identity follows. □

For L^2 , this is convergence in norm; pointwise convergence is a separate question.

The Fourier example

Example (verbatim). In $L_2(-\pi, \pi)$, the collection (or any subset thereof)

$$x_n = \frac{1}{\sqrt{2\pi}} e^{int}, \quad n = 0, \pm 1, \dots$$

is an orthonormal set of vectors. **Proof.** For any $n \in \mathbb{Z}$,

$$\int_{-\pi}^{\pi} x_n \overline{x_n} dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{int} \overline{e^{int}} dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} \|e^{int}\|^2 dt = 1,$$

functional-analysis/functional-analysis.tex, “Orthogonal projections”.

The Fourier example: proof continued

Proof (continued, verbatim). and if m is another integer,

$$\int_{-\pi}^{\pi} x_n \overline{x_m} dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{int} \overline{e^{imt}} dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(n-m)it} dt = \frac{1}{2\pi} \left[\frac{e^{(n-m)it}}{(n-m)i} \right]_{-\pi}^{\pi} = 0.$$

In this last calculation, “another integer” means $m \neq n$.

This proves orthonormality. To obtain a basis, we must also prove density.

`functional-analysis/functional-analysis.tex`, “Orthogonal projections”.

The real-valued version

Example (verbatim). If we restrict our attention to only real-valued functions that are square-integrable on the interval $[-\pi, \pi]$, then the collection (or any subset thereof)

$$\frac{1}{\sqrt{2\pi}}, \frac{1}{\sqrt{\pi}} \cos t, \frac{1}{\sqrt{\pi}} \cos 2t, \dots$$
$$\frac{1}{\sqrt{\pi}} \sin t, \frac{1}{\sqrt{\pi}} \sin 2t, \dots$$

is an orthonormal set. We use the complex exponential version in the rest of the lecture.

`functional-analysis/functional-analysis.tex`, “Orthogonal projections”.

From an orthonormal system to a basis

The same integral over $(0, 2\pi]$ shows that

$$\phi_n(t) = \frac{e^{int}}{\sqrt{2\pi}}, \quad n \in \mathbb{Z},$$

is orthonormal in $L^2(0, 2\pi]$.

Completeness. Finite sums of e^{int} are trigonometric polynomials. They form a subalgebra of $C(S^1, \mathbb{C})$, contain constants, separate points, and are closed under complex conjugation.

The complex Stone–Weierstrass theorem therefore makes them uniformly dense in $C(S^1, \mathbb{C})$. Continuous functions are dense in $L^2(S^1)$, and $\|h\|_2 \leq \sqrt{2\pi}\|h\|_\infty$. Thus their span is dense in L^2 . □

This is the completeness argument in Verbitsky, Lecture 12, pp. 2–4.

The Fourier expansion on the circle

By the Hilbert-space expansion theorem,

$$f(t) = \sum_{n \in \mathbb{Z}} a_n e^{int}, \quad a_n = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-int} dt.$$

The sum converges in L^2 .

From now on we use the normalized measure $dt/(2\pi)$. Then e^{int} itself has norm one, and the coefficients are simply its inner products with f .

An L^2 function on the interval defines a function on the circle up to a set of measure zero. Smooth functions on the circle correspond to smooth periodic functions on $\mathbb{R}$.

The Fourier basis on the torus

Let

$$E = \mathbb{C}/(2\pi\mathbb{Z} + 2\pi i\mathbb{Z}), \quad z = x + iy.$$

Functions on E are periodic in both x and y . For the measure $dx\,dy/(2\pi)^2$, put

$$e_{n,m}(x,y) = e^{inx+imy}, \quad (n,m) \in \mathbb{Z}^2.$$

Proof that this is a basis. Orthogonality follows by multiplying the two one-variable integrals:

$$\langle e_{n,m}, e_{r,s} \rangle = \delta_{nr}\delta_{ms}.$$

Their span is again a conjugation-closed algebra containing constants and separating points of E . Stone–Weierstrass and the density of $C(E)$ in $L^2(E)$ prove completeness. □

Fourier modes of functions and forms

Every $f \in L^2(E)$ has the expansion

$$f = \sum_{n,m \in \mathbb{Z}} a_{n,m} e_{n,m}, \quad a_{n,m} = \frac{1}{(2\pi)^2} \int_E f(x,y) e^{-inx - imy} dx dy.$$

The term $a_{n,m} e_{n,m}$ is its (n,m) -**Fourier mode**. There are two integer indices because there are two real coordinates.

Expand the coefficients of a differential form in the same way:

$$\eta = g d\bar{z} = \sum_{n,m} a_{n,m} e_{n,m} d\bar{z}.$$

More generally, a mode is $e_{n,m} \alpha$ with α a constant form. The **zero mode** is $(n,m) = (0,0)$: its coefficients are constant. The frequency (n,m) and the form type (p,q) are different indices.

The exterior derivative preserves modes

Claim. Differentiating a mode keeps the same frequency.

Proof. Direct differentiation gives

$$de_{n,m} = e_{n,m}(in\,dx + im\,dy).$$

If α is a constant form, then $d\alpha = 0$, so

$$d(e_{n,m}\alpha) = e_{n,m}(in\,dx + im\,dy) \wedge \alpha.$$

Every coefficient is a multiple of the same basis function $e_{n,m}$. □

The form degree changes; the pair (n, m) does not.

The Dolbeault operator preserves modes

Our conventions are

$$z = x + iy, \quad I\partial_x = \partial_y, \quad D_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y), \quad \bar{\partial} = d\bar{z} \wedge D_{\bar{z}}.$$

Proof. Since $\partial_x e_{n,m} = i n e_{n,m}$ and $\partial_y e_{n,m} = i m e_{n,m}$,

$$\bar{\partial}(e_{n,m}\alpha) = \frac{in - m}{2} e_{n,m} d\bar{z} \wedge \alpha \quad (\alpha \text{ constant}).$$

Thus $\bar{\partial}$ also preserves each frequency. □

By uniqueness of Fourier coefficients, if $\bar{\partial}\eta = 0$, each mode of η is $\bar{\partial}$ -closed. We can seek primitives mode by mode. For smooth forms, termwise differentiation is justified below.

Complex Cartan's formula

Let $X = \partial_x$ and $IX = \partial_y$ be translation fields on E . Contraction is denoted by ι_X ; on a one-form, $\iota_X \eta = \eta(X)$. Lie derivatives differentiate coefficients.

The real Cartan formula is $d\iota_X + \iota_X d = \mathcal{L}_X$. Its complex versions are

$$\bar{\partial}\iota_X + \iota_X \bar{\partial} = \frac{1}{2}(\mathcal{L}_X + i\mathcal{L}_{IX}),$$

$$\partial\iota_X + \iota_X \partial = \frac{1}{2}(\mathcal{L}_X - i\mathcal{L}_{IX}).$$

With $z = x + iy$ and $IX = \partial_y$, the plus sign belongs to $\bar{\partial}$.

Compare Verbitsky, Lecture 12, p. 13; signs here follow the displayed coordinate conventions.

Proof of complex Cartan for translations

For every form ω , the contraction rule gives

$$\iota_X(d\bar{z} \wedge \omega) = \underbrace{d\bar{z}(X)}_1 \omega - d\bar{z} \wedge \iota_X \omega.$$

Also $D_{\bar{z}}$ commutes with ι_X , since X is constant. Therefore

$$\begin{aligned}(\bar{\partial} \iota_X + \iota_X \bar{\partial})\omega &= d\bar{z} \wedge \iota_X D_{\bar{z}}\omega + \iota_X(d\bar{z} \wedge D_{\bar{z}}\omega) \\ &= D_{\bar{z}}\omega = \frac{1}{2}(\mathcal{L}_X + i\mathcal{L}_{IX})\omega.\end{aligned}$$

Replacing $d\bar{z}, D_{\bar{z}}$ by $dz, \frac{1}{2}(\partial_x - i\partial_y)$ proves the formula for ∂ . Adding the two identities recovers real Cartan. □

A mode is an eigenvector

Let $\eta_{n,m} = e_{n,m}\alpha$, with α constant. Since translations leave α fixed,

$$\mathcal{L}_X \eta_{n,m} = in\eta_{n,m}, \quad \mathcal{L}_{IX} \eta_{n,m} = im\eta_{n,m}.$$

For the combination with a minus sign,

$$(\mathcal{L}_X - i\mathcal{L}_{IX})\eta_{n,m} = (in + m)\eta_{n,m}.$$

The combination used for $\bar{\partial}$ is

$$(\mathcal{L}_X + i\mathcal{L}_{IX})\eta_{n,m} = \lambda_{n,m}\eta_{n,m}, \quad \lambda_{n,m} = in - m.$$

For integer n, m , $\lambda_{n,m} = 0$ exactly when $(n, m) = (0, 0)$.

A closed nonzero mode is exact

Let $\eta_{n,m} = a_{n,m} e_{n,m} d\bar{z}$ and $(n, m) \neq (0, 0)$. Suppose $\bar{\partial}\eta_{n,m} = 0$ (automatic for $(0, 1)$ -forms).

Proof. Complex Cartan gives

$$\frac{\lambda_{n,m}}{2} \eta_{n,m} = \bar{\partial} \iota_X \eta_{n,m} + \iota_X \bar{\partial} \eta_{n,m} = \bar{\partial} \iota_X \eta_{n,m}.$$

Since $\lambda_{n,m} \neq 0$,

$$\eta_{n,m} = \bar{\partial} f_{n,m}, \quad f_{n,m} = \frac{2}{\lambda_{n,m}} \iota_X \eta_{n,m} = \frac{2a_{n,m}}{in - m} e_{n,m}. \quad \square$$

The conclusion is that the mode is *exact*, not that it is constant.

Adding the primitives

Let $\eta = g d\bar{z}$ be a smooth $(0, 1)$ -form on E , with

$$g = \sum_{n,m} a_{n,m} e_{n,m}.$$

Separate its constant coefficient and define

$$c = a_{0,0} = \frac{1}{(2\pi)^2} \int_E g dx dy, \quad f = \sum_{(n,m) \neq (0,0)} \frac{2a_{n,m}}{in - m} e_{n,m}.$$

We will justify that f is smooth and can be differentiated term by term. Then

$$\bar{\partial} f = (g - c) d\bar{z}, \quad \boxed{\eta = c d\bar{z} + \bar{\partial} f.}$$

Only the zero mode was excluded when dividing by $in - m$.

Why the sums and their derivatives converge

Put $\Delta = \partial_x^2 + \partial_y^2$ and $r^2 = n^2 + m^2$. Repeated integration by parts on the torus gives, for every $N \geq 0$,

$$(1 + r^2)^N a_{n,m} = \widehat{(1 - \Delta)^N g}(n, m), \quad |a_{n,m}| \leq C_N (1 + r^2)^{-N}.$$

There are no boundary terms because g is periodic. For $r \geq 1$, $|in - m| = r$, so the coefficients $b_{n,m} = 2a_{n,m}/(in - m)$ have the same rapid decay.

A derivative of order k multiplies a mode by a number of modulus at most r^k . Choose $N > (k + 2)/2$; then

$$\sum_{(n,m) \neq (0,0)} r^k (1 + r^2)^{-N} < \infty.$$

Thus all derivative series converge uniformly, for both g and f . Their L^2 expansions identify the limits, and $\bar{\partial}f = (g - c)d\bar{z}$. □

The constant part is exactly the cohomology

We have proved

$$\eta = c d\bar{z} + \bar{\partial}f \implies [\eta] = c[d\bar{z}].$$

Why does a nonzero constant survive? For every smooth periodic h ,

$$\frac{1}{(2\pi)^2} \int_E \frac{\partial h}{\partial \bar{z}} dx dy = \frac{1}{2(2\pi)^2} \int_E (h_x + ih_y) dx dy = 0.$$

Thus $c d\bar{z} = \bar{\partial}h$ forces $c = 0$. The coefficient c is uniquely determined by the cohomology class. Consequently

$$H_{\bar{\partial}}^{0,1}(E) = \mathbb{C}[d\bar{z}]. \quad \square$$

All Dolbeault groups of the elliptic curve

(p, q)	constant representative	$H_{\bar{\partial}}^{p,q}(E)$
$(0, 0)$	1	$\mathbb{C}$
$(1, 0)$	dz	$\mathbb{C}$
$(0, 1)$	$d\bar{z}$	$\mathbb{C}$
$(1, 1)$	$dz \wedge d\bar{z}$	$\mathbb{C}$

Proof of the remaining cases. If $\bar{\partial}h = 0$, every nonzero Fourier coefficient is zero, since $\bar{\partial}e_{n,m} = (in - m)e_{n,m} d\bar{z}/2$. Thus h is constant; the same applies to a closed form $h dz$.

For $g dz \wedge d\bar{z}$, write $g = c + D_{\bar{z}}f$. Then

$$g dz \wedge d\bar{z} = c dz \wedge d\bar{z} + \bar{\partial}(-f dz).$$

The averaging argument again shows that a nonzero constant is not exact. Hence constant forms represent cohomology uniquely. □

Put the disk inside a large fundamental domain

Let $\eta = g d\bar{z}$ be smooth on an open neighbourhood U of $\overline{D}$. Choose L large enough that

$$\overline{D} \subset Q^\circ, \quad Q = [-L/2, L/2] \times [-L/2, L/2], \quad E_L = \mathbb{C}/(L\mathbb{Z} + iL\mathbb{Z}).$$

The quotient map embeds Q° into E_L . Thus the disk fits inside the elliptic curve without identifying its points.

The Fourier calculation works with period L :

$$e_{n,m}^L = e^{2\pi i(nx+my)/L}, \quad \bar{\partial} e_{n,m}^L = \frac{\pi}{L}(in - m)e_{n,m}^L d\bar{z}.$$

The same nonzero eigenvalues are invertible, so the same decomposition holds.

Extend the form using a cutoff

Choose $\chi \in C_c^\infty(U \cap Q^\circ)$ with $\chi = 1$ near $\overline{D}$. Define on Q

$$\tilde{\eta} = \chi\eta,$$

extending by zero outside U . It vanishes near the edges of Q , so periodic extension defines a smooth $(0,1)$ -form on E_L .

It agrees with η on D . It is also $\bar{\partial}$ -closed:

$$\bar{\partial}\tilde{\eta} = \frac{\partial(\chi g)}{\partial\bar{z}} d\bar{z} \wedge d\bar{z} = 0.$$

This automatic closedness is where complex dimension one is used.

The last step: the constant form is exact on the disk

On the elliptic curve, the preceding argument gives

$$\tilde{\eta} = c d\bar{z} + \bar{\partial}f \quad (c \in \mathbb{C}, f \in C^\infty(E_L)).$$

Restrict to the disk. There $\bar{z}$ is a well-defined function and

$$d\bar{z} = \bar{\partial}\bar{z}.$$

Consequently

$$\eta = \bar{\partial}(c\bar{z} + f|_D).$$

Thus $u = c\bar{z} + f|_D$ is the required smooth primitive. □

On the torus, $\bar{z}$ is not a function: it changes under lattice translations. On the disk it is available, and removes the constant term.

The one-variable lemma in every positive degree

For type $(0, 1)$, we have just proved local solvability of $\bar{\partial}u = g d\bar{z}$.

For type $(1, 1)$, write $\eta = g dz \wedge d\bar{z}$ and use the same u :

$$\bar{\partial}(-u dz) = -\bar{\partial}u \wedge dz = g dz \wedge d\bar{z} = \eta.$$

There are no forms with $q > 1$ in one complex dimension. Every germ has a representative near a closed smaller disk, so the proof establishes the local statement for all smooth forms.

Hence, for $p = 0, 1$, the sequence of sheaves

$$0 \longrightarrow \Omega^p \longrightarrow \mathcal{A}^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1} \longrightarrow 0$$

is exact. This is the one-dimensional input to the Dolbeault resolution.

Sources

- Misha Verbitsky, *Hodge theory*, IMPA, 2025: Lecture 11, Lecture 12, and Lecture 13.
- Daniel González Casanova Azuela, `de-rham-dolbeault` notes: acyclic resolutions, fine sheaves, and the proof of Dolbeault's theorem.
- Daniel González Casanova Azuela, `functional-analysis/functional-analysis.tex`: “Orthogonal projections”, the examples beginning “the collection (or any subset thereof)”. The quoted text and formulas are reproduced verbatim.